# Equilibrium: find the missing balance

**LEARNING TARGET** I can find a missing force when acceleration is zero.

## A · KEY VOCABULARY

| | |
|---|---|
| **Equilibrium** | Acceleration is zero. The forces sum to zero in every direction. |
| **Two kinds** | At rest *or* moving at constant velocity. |
| **Missing force** | Whatever is needed to make each direction balance. |
| **Careful** | Zero velocity for an instant does not guarantee equilibrium. |

## B · KEY CONDITION

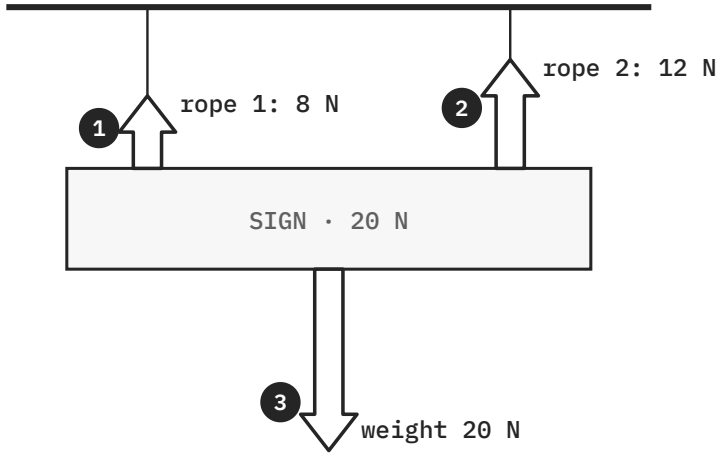
$$\Sigma F = 0$$

up = down
right = left

Σ "sum of"
force newtons (N)

## C · COLOR IT · SUPPORTS MUST TOTAL THE WEIGHT

*8 N + 12 N = 20 N.*




☐ **1 · ORANGE** Rope 1 tension
☐ **2 · ORANGE** Rope 2 tension
☐ **3 · ORANGE** Weight (W)

## D · WORKED EXAMPLE

A 40 N sign hangs at rest from two vertical ropes. One tension is 15 N. Find the other.

| GIVEN | EQUATION | WORK | ANSWER |
|---|---|---|---|
| weight 40 N<br>rope 1 = 15 N<br>at rest | up = down | $15 + T_2 = 40$<br>$T_2 = 40 - 15$ | **25 N up** |

## E · THE 2026-XJ THREAD

DAY 18 OF 22 · NEW TOOL: EQUILIBRIUM

If the forces on XJ balanced, its acceleration would be zero and its velocity would stay constant. Analysts check for balance before assuming steady motion.

*Unit question: How fast is it moving, and where will it arrive?*

**UNIT 1 COLOR CODE** ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else